

Span2Type

Improved Open-World Named Entity Recognition

Motivation & Problem Definition

Khoa Nguyen

April 27, 2026

- 1 Motivation
- 2 Five Challenges
- 3 Problem Definition

Named Entity Recognition Powers Everything

NER underpins:

- Information extraction
- Knowledge graph construction
- Question answering
- Biomedical text mining
- Document understanding

State of the Art

BERT-based NER achieves

near-human performance

on CoNLL-2003 & OntoNotes

Open Question

If NER works so well on benchmarks —
why can't we just deploy it everywhere?

The Hidden Assumption Breaking Real-World NER

Benchmark Setting	Real World
Types fixed in advance	Types unknown , evolve over time
Large labeled training set	Annotation: expensive, slow, expert-needed
One domain, one ontology	New domains emerge constantly
ORG / PER / LOC / MISC	Protein / Subsystem / Regulatory Mechanism

Five Open Problems in Open-World NER

- C1** Span boundaries break under domain shift
- C2** The type inventory is unknown
- C3** Cluster IDs are not interpretable
- C4** Full fine-tuning is too expensive per domain
- C5** Standard metrics fail for open-vocabulary names

Each challenge motivates one targeted design decision in Span2Type.

C1 — Span Boundaries Break Under Domain Shift

Source (newswire):

``Germany defeated Brazil 2-1.``

✓ Simple nouns – BIO handles fine

Target (CrossNER-AI):

``We apply **support vector machine**
to image classification.``

× 3-token span – BIO often splits it

Root cause: BIO sees **one signal** only

BIO sequential per-token tags

SE start / end boundary tokens

TB adjacent token co-membership

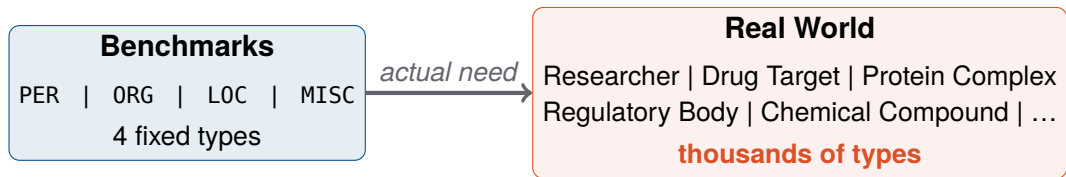
Key Takeaway

All three views exist.

OWNER uses only BIO.

60.5 F1 on CrossNER-AI
— lowest of all 5 domains.

C2 — The Type Inventory Is Unknown



- Real-world types number in the **thousands** (Choi et al., 2018)
- Supervised NER only classifies *trained* types
- New types are invisible to closed-world systems

GLiNER / UniversalNER still fail here

“Zero-shot” NER still requires a human to **supply type names at inference**.

True open-world: the system **discovers types by itself**.

C3 — Cluster IDs Are Not Entity Types

Prior work stops here:

```
``Albert Einstein'' → Cluster 3  
``Marie Curie''    → Cluster 3  
``Watson''         → Cluster 7
```

What is Cluster 3?

Cannot validate. Cannot deploy.

Span2Type goes further:

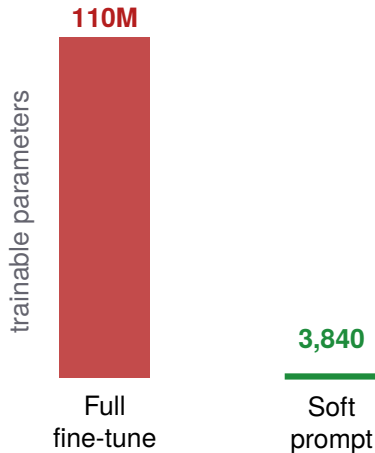
```
Cluster 3 → ``Physicist''  
Cluster 7 → ``Fictional Character''
```

- ✓ Expert-validatable
- ✓ Knowledge-base ready
- ✓ Downstream-usable

Key Takeaway

Naming is **not cosmetic** — it is what makes unsupervised clusters **actionable**.

C4 — Full Fine-Tuning Is Too Expensive per Domain



Full fine-tuning risks:

- Costly per new domain
- Catastrophic forgetting of pre-trained

Solution: two-stage schedule

Stage 1: backbone **frozen**, prompt warms

C5 — Standard Metrics Fail for Open-Vocabulary Names

Exact match is misleading:

Gold: "musicalartist"
Output: "musician"
Exact-match score: 0
Semantic reality: correct

Gold: "researcher"
Output: "physicist"
Exact-match score: 0
Semantic reality: close hyponym

BERTScore-F1 calibration:

Prediction quality	Score
Exact copy	1.00
Synonym / hyponym	≈ 0.95
Related but off	≈ 0.90
Completely wrong	≈ 0.86

Key Takeaway

Signal lives in $[0.86, 1.0]$.
Reproducible & scalable —
human eval infeasible at this scale.

Challenges → Contributions

	Challenge	Our Answer
C1	Brittle single-view span detection	DTrans: BIO + SE + TB union
C2	Unknown type inventory	BIC-guided auto k^* selection
C3	Cluster IDs not interpretable	MMR exemplar selection + LLM naming
C4	Fine-tuning cost per domain	Two-stage soft prompt (3,840 params)
C5	No metric for open-vocab names	BERTScore-F1 as semantic proxy

Every methodology design choice traces back to a row in this table.

Formal Problem Statement

Input

An **unannotated** target-domain corpus $\mathcal{D}_T$

No gold entity spans | no type labels | no ontology

Goal — Three Outputs

- 1 **Span detection:** identify all entity mention spans in $\mathcal{D}_T$
- 2 **Entity typing:** group spans into coherent semantic clusters (unsupervised, *k* unknown)
- 3 **Cluster naming:** assign a concise, human-readable label to each cluster

Evaluation constraint

Gold CrossNER labels used **strictly for evaluation** — never seen during training, clustering, or naming.

Open-World vs. Closed-World NER

Property	Closed-World NER	Open-World (Span2Type)
Type set at train time	Fixed, known	Unknown
Labeled target data	Required	Not used
Output	Span + predefined type	Span + discovered type + name
Handles new types?	No	Yes
Example types	PER, ORG, LOC	Physicist, Drug Compound

How We Measure Success

Stage	Metric	What It Measures
Entity Detection	Span F1 (exact)	Precision / Recall of span boundaries
Entity Typing	AMI	Cluster quality vs. gold partition — no label mapping needed
Cluster Naming	BERTScore-F1	Semantic similarity: predicted vs. gold type name

Source

CoNLL-2003 (4 types, newswire)
Detector training only

Target

CrossNER — 5 domains
AI | Literature | Music | Politics | Science
9–17 gold types each

End of Section

Motivation & Problem Definition

Next: Methodology — Three Contributions